

Duncan O'Brien

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Employment

2027 – date	Research Fellow, University of Sheffield, UK
2024 – 2026	Senior Research Associate, University of Bristol, UK
2019 – 2020	Farmer, Manor Farm, Dorset, UK
2018	Research Technician, Smithsonian Institute, The Bahamas

Education

2020 – 2024	PhD in Quantitative Ecology, University of Bristol, UK
2018 – 2019	MSc in Tropical Marine Biology, University of Essex UK
2015 – 2018	BA Biological Sciences, University of Oxford, UK

Publications

I have published **18** peer reviewed papers, 9 of which are first author. Based upon Google Scholar metrics, these have been **cited 260 times** (as of July 2026, **h-index = 9**). First and senior author publications are indicated by an asterisk (*).

18. Rodríguez-Hernandez D, Fischer F, **O'Brien DA**, et al. (2026) Limited predictability of tree-level responses to drought across European forests. [Forest Ecology and Management](#).

17*. Capdevila P, **O'Brien DA**, et al. (Co-first author) (2026) Halting predicted vertebrate declines requires tackling multiple drivers of biodiversity loss. [Science Advances](#).

16*. **O'Brien DA**, Clements CF (2025) Stability metrics behave predictably across data qualities but are sensitive to community size. [Biological Conservation](#).

15. Cerini F, ... **O'Brien DA**, et al. (2025) Multivariate signals of population collapse in a high-throughput ecological experiment. [Ecology](#).

14. Campbell JE, ... **O'Brien DA**, et al. (2024) Herbivore effects increase with latitude across the extent of a foundational seagrass. [Nature Ecology & Evolution](#).

13*. **O'Brien DA**, et al. (2023) Early warning signals have limited applicability to empirical lake data. [Nature Communications](#).

12*. **O'Brien DA**, et al. (2023) Planktonic functional diversity changes in synchrony with lake ecosystem state. [Global Change Biology](#).

11*. **O'Brien DA**, et al. (2023) EWSmethods: an R package to forecast tipping points at the community level using early warning signals, resilience measures, and machine learning models. [Ecography](#).

10*. Mason HD, Rose E, Gonzalez JE, **O'Brien DA** (2023) Nocturnal surveys of lined seahorses reveal increased densities and seasonal recruitment patterns. [Ecology and Evolution](#).

9. Besson M, ... **O'Brien DA**, et al. (2023) Anemone bleaching impacts the larval recruitment success of an anemone-associated fish. [Coral Reefs](#).

8. Cerini F, **O'Brien DA**, et al. (2023) Phenotypic response to different predator strategies can be mediated by temperature. [Ecology and Evolution](#).

7. Fourqurean JW, ... **O'Brien DA**, et al. (2023) Seagrass abundance predicts surficial soil organic carbon stocks across the range of *Thalassia testudinum* in the western North Atlantic. [Estuaries and Coasts](#).

6. Wolfe E, ... **O'Brien DA**, et al. (2023) Spatiotemporal thermal variation drives diversity trends in experimental landscapes. [Journal of Animal Ecology](#).

5*. **O'Brien DA**, Clements CF (2022) Early warning signal reliability varies with COVID-19 waves. [Biology Letters](#).

4*. **O'Brien DA**, et al. (2021) An experimental assessment of social tolerance and den ecology in a high-density octopus population. [Marine Biology](#).

- 3*. **O'Brien DA**, et al. (2021) Nocturnal surveys reveal novel insights into the community dynamics of an anchialine ecosystem from The Bahamas. [Estuarine, Coastal and Shelf Science](#).
2. O'Shea O, ... **O'Brien DA**, et al. (2021) Evidence and description of a nursery habitat for the recently reclassified stingray *Styracura schmardae* from The Bahamas. [Marine Ecology Progress Series](#).
- 1*. **O'Brien DA**, et al. (2020) Drivers of octopus abundance and density in an anchialine lake: A 30 year comparison. [Journal of Experimental Marine Biology and Ecology](#).

Funding

2026 – date	<i>Ecological time lags and their importance for causal inference</i> Commission for the Exhibition of 1851 – Fellowship	£171,100
2024 – 2026	<i>MULTI-STRESS: Quantifying the impacts of multiple stressors in multiple dimensions to improve ecological forecasting</i> NERC: Pushing the Frontiers (NE/Z000130/1) – Named postdoc and author	£674,036
2020 – 2024	<i>Indicators of regime shifts in freshwater ecosystems</i> NERC: PhD Studentship (grant number 2439776) – Co-I	£56,855

Awards

2025	Freshwater Biological Association best thesis
2025	European Federation for Freshwater Sciences best thesis - Runner up
2022	British Ecological Society best student poster
2019	University of Essex Marine Biology prize for best marine biology focussed dissertation

Scientific Presentations

Invited talks

2025	<i>Early warning signals have limited applicability to empirical lake data</i> 14 th Symposium for European Freshwater Sciences, Bolu, Turkey
2025	<i>Causal inference corroborates presumed drivers of seabird declines</i> SEAPOP, Norwegian Institute for Nature Research, Norway
2025	<i>Timing and the forecast-ability of seabird dynamics</i> Avian Ecology Group at the University of Liverpool (AEGUL), Liverpool, UK
2022	<i>Resilience as an interdisciplinary concept</i> Research Without Borders, Bristol, UK
2020	<i>Octopus abundance in an anchialine lake from The Bahamas</i> Bahamas Natural History Conference, Nassau, The Bahamas

Contributed talks

2024	<i>The Timeline to Collapse suggests terminal decline in North Atlantic seabirds</i> British Ecological Society annual meeting, Liverpool, UK
2023	<i>Resilience metrics are robust across data qualities but sensitive to community size</i> British Ecological Society annual meeting, Belfast, UK
2022	<i>Does functional diversity lead changes in ecosystem state?</i> International Centre for Theoretical Science, Bangalore, India
2022	<i>Identifying regime shifts using early warning signals: univariate and multivariate options</i> British Ecological Society annual meeting, Edinburgh, UK
2021	<i>Planktonic functional diversity changes in synchrony with ecosystem state</i> British Ecological Society annual meeting, Liverpool, UK

Teaching Experience

2025 – 2026	Co-supervise 1 PhD and 1 MRes project at Sorbonne University, France
2025	Guest lecture: <i>Causal inference and its application to seabird ecology</i> . Norwegian Institute for Nature Research, Norway
2023	Written software tutorials for the <i>EWSmethods</i> R package
2020 – 2023	Teaching assistant MSc Bioinformatics, Ecology and Conservation & Evolutionary Biology at University of Bristol, UK
2019	Leader postgraduate marine ecology field course at Centre for Ocean Research and Education, The Bahamas

Service

Committee membership	Steering committee – European Ecological Forecasting Initiative
Software maintainer	<i>EWSmethods</i> R package (https://cran.r-project.org/package=EWSmethods)
Reviewership	Advanced Science, Biology Letters (x2), Ecography (x3), Ecology (x2), Journal of Animal Ecology (x6), Journal of Ecology, Limnology and Oceanography Letters (x4), Oikos, Proceedings B, Nature Communications